مدينة زويل للعلوم والتكنولوجيا
Zewail City of Science and Technology

Spacecraft Communication Systems (SPC 535)

Dr. Samy S. Soliman
Eng. Eman Abdelnaby
Eng. Aya Shehatta

Project 1
Analog Communications Project

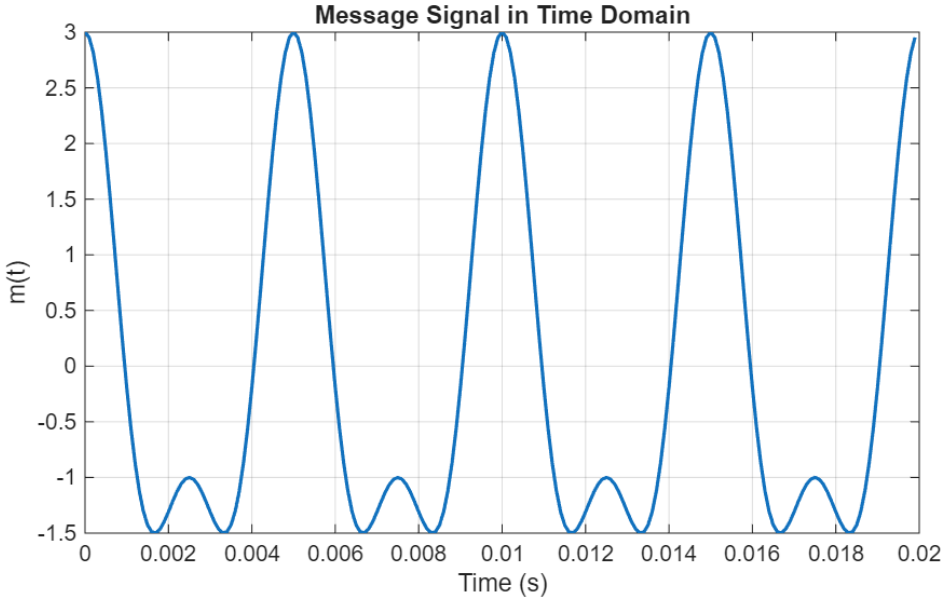
Asmaa Sapry **202201338**

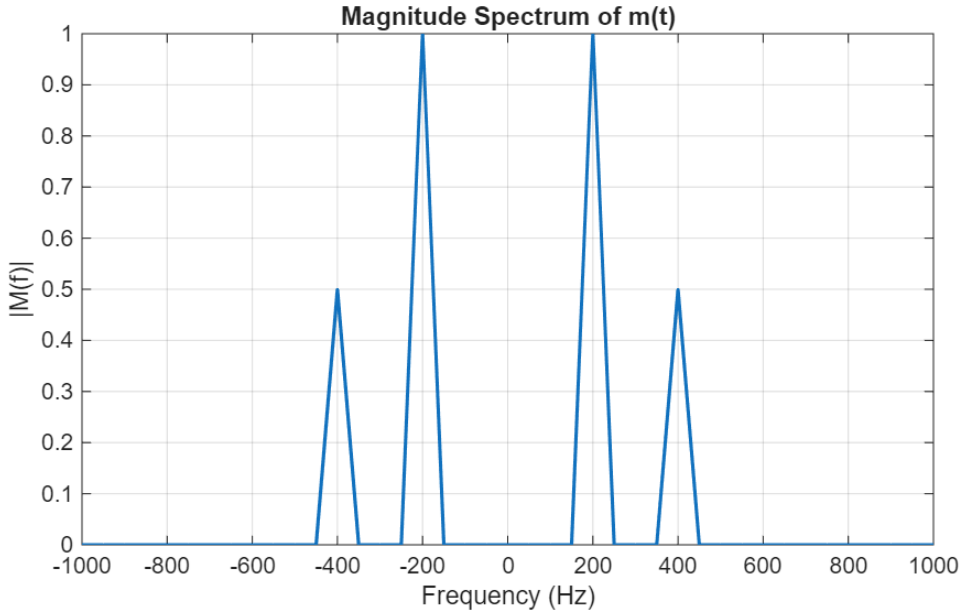
Raed Asham **202200272**

Date of Submission: **3/18/2026**

Part A: Amplitude Modulation (AM)

A1: Message Signal

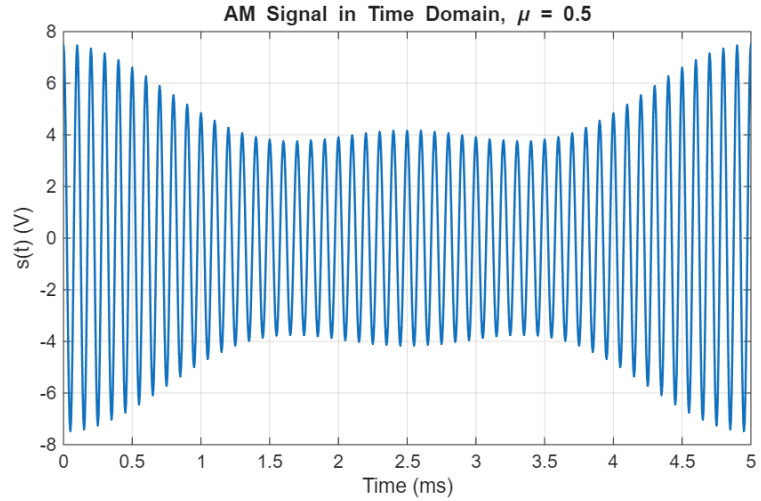


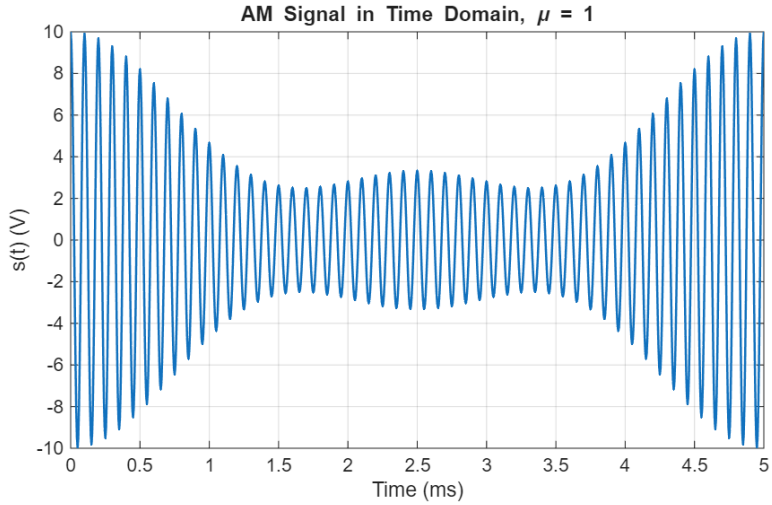


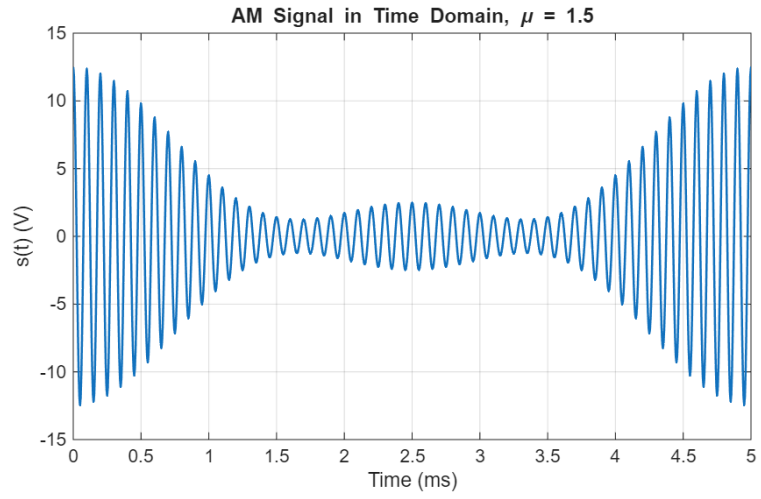
Observations:

- Frequency components: $200\ Hz$ and $400\ Hz$
- Message bandwidth: $B_m = 400\ Hz$

A2: Conventional AM (DSB-LC)



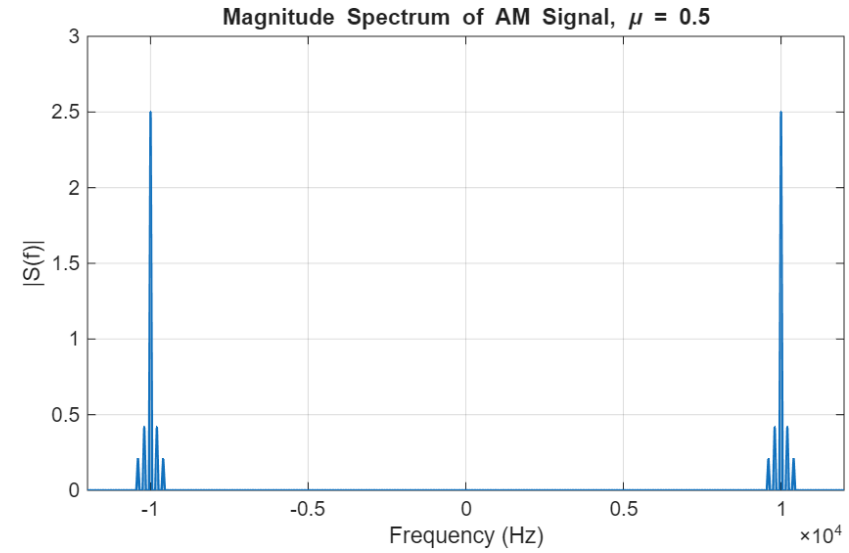


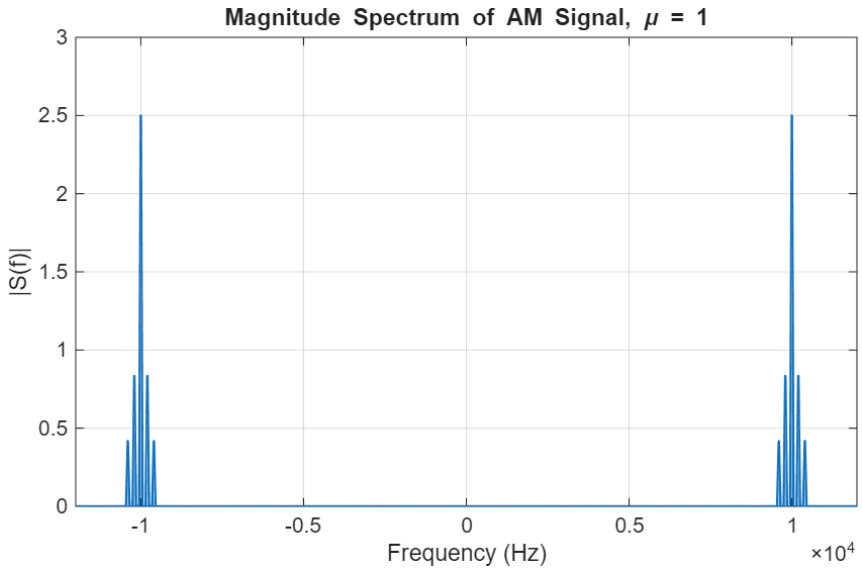


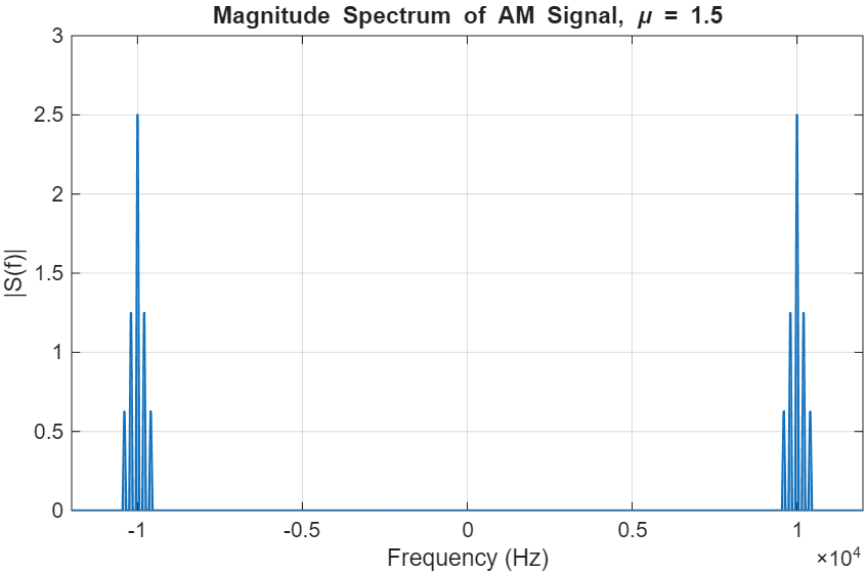
Comments:

- $\mu = .5$: under-modulation.
- $\mu = 1$: critical modulation.
- $\mu = 1.5$: formally above 1, so it is usually called over-modulation.

❖ Plotting the frequency spectrum







❖ Bandwidth Estimation and Power Efficiency (from MatLab Code Output)

Command Window

```
Maximum of m(t) = 3.0000
Message bandwidth Bm = 400.0 Hz
Transmission bandwidth = 800.0 Hz

Carrier power Pc = 12.500000 W (for 1-ohm load)
Normalized message power Pmn = 0.277778

mu = 0.5
  Theoretical efficiency = 0.064935 (6.49%)
  Numerical efficiency   = 0.064935 (6.49%)
  Total power            = 13.368056 W
  Sideband power         = 0.868056 W

mu = 1.0
  Theoretical efficiency = 0.217391 (21.74%)
  Numerical efficiency   = 0.217391 (21.74%)
  Total power            = 15.972222 W
  Sideband power         = 3.472222 W

mu = 1.5
  Theoretical efficiency = 0.384615 (38.46%)
  Numerical efficiency   = 0.384615 (38.46%)
  Total power            = 20.312500 W
  Sideband power         = 7.812500 W

Minimum of normalized message mn(t) = -0.499998
mu = 0.5 --> minimum envelope factor [1 + mu*mn(t)] = 0.750001
mu = 1.0 --> minimum envelope factor [1 + mu*mn(t)] = 0.500002
mu = 1.5 --> minimum envelope factor [1 + mu*mn(t)] = 0.250003
```

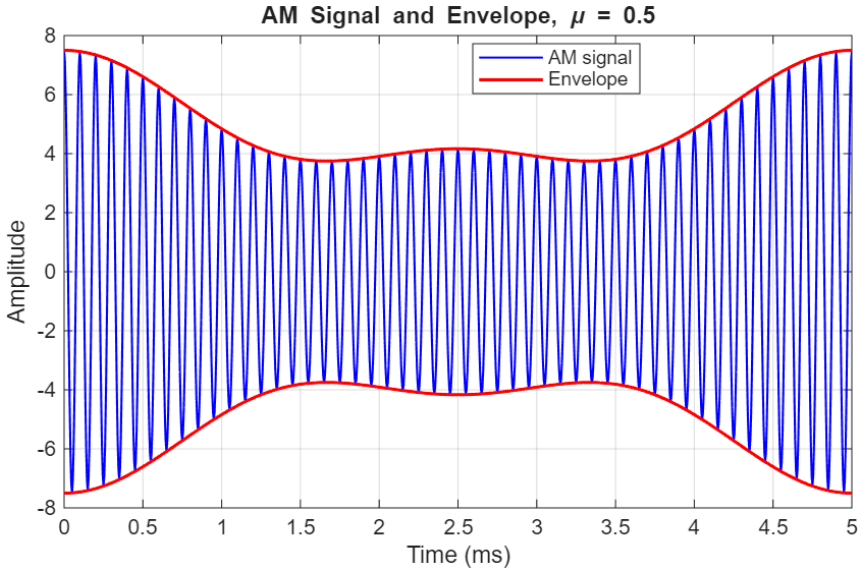
Comments:

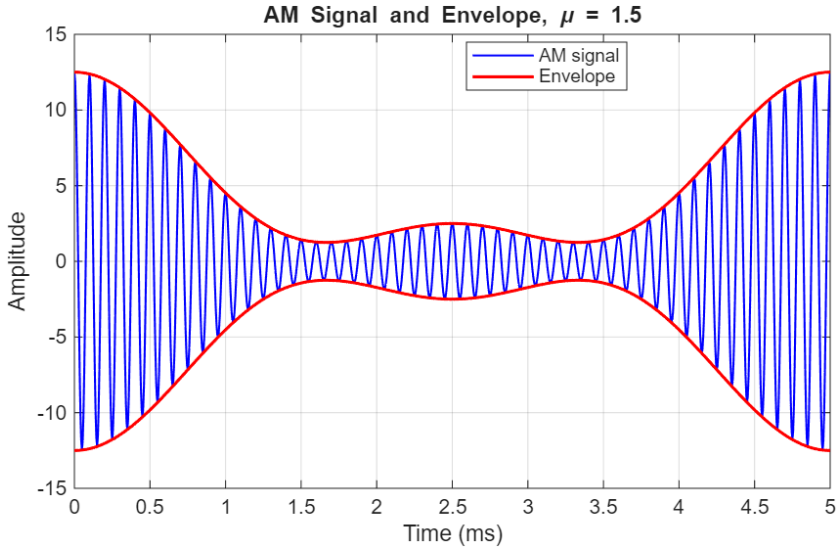
- Power Efficiency is calculated using the following equations:

$$P_c = \frac{A_c^2}{2}, \eta = \frac{P_{SB}}{P_T}, \eta_{\text{th}} = \frac{\mu^2 P_{m_n}}{1 + \mu^2 P_{m_n}}$$

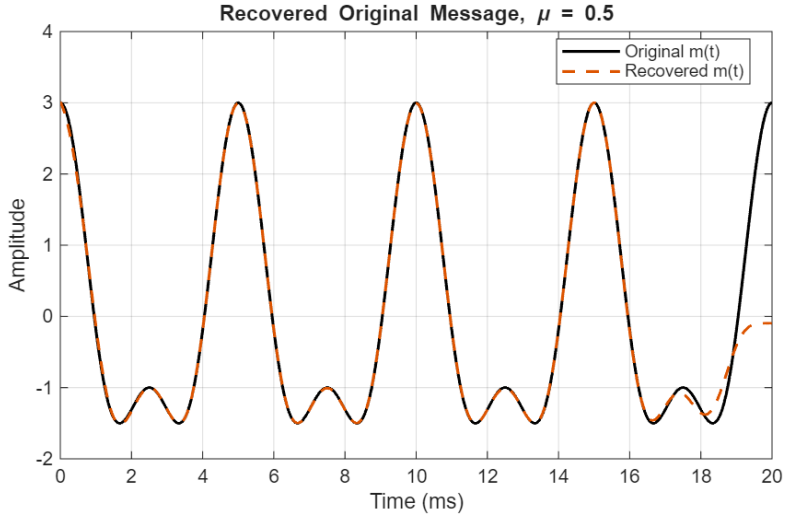
A3: Envelope Detection

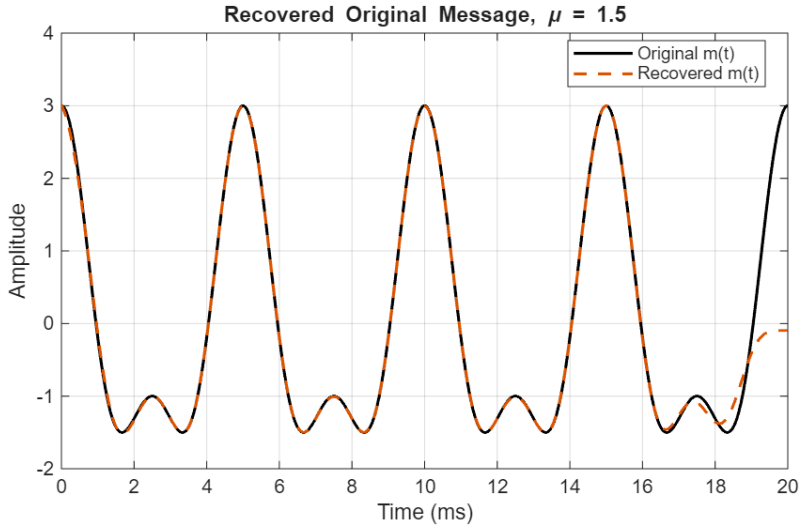
❖ Envelope Detection





❖ Recovered Signals





Observations:

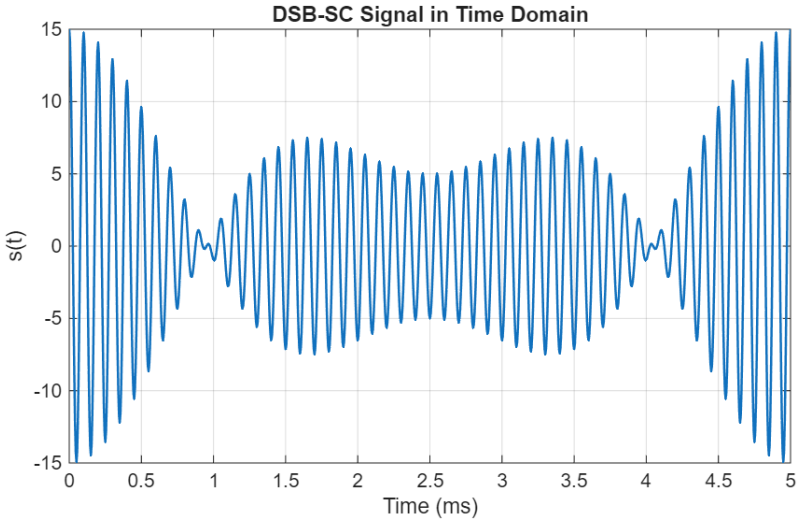
- For $\mu = 0.5$
  - This is under-modulation
  - The envelope follows the message well
  - The recovered signal should match the original message closely
  - Distortion is very small
- For $\mu = 1.5$
  - This is over-modulation because $\mu > 1$
  - However, for this specific message normalization, the minimum of $m_n(t)$ is not -1, it is about -0.5.
  - Hence, the envelope does not cross zero for this particular signal, so the detector still works much better than the usual over-modulated case.

Comments on Effects of Over-Modulation:

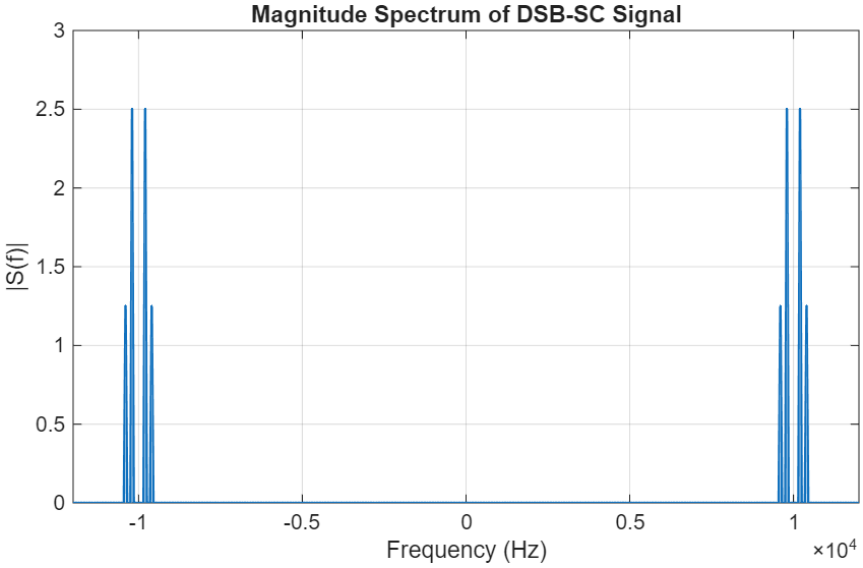
- Over-modulation causes:
  - Envelope crossing or inversion
  - Distortion in envelope detection
  - Failure to recover the original message accurately
- Although $\mu = 1.5$ exceeds the standard AM limit and is normally classified as over-modulation, the specific normalized message used here does not drive the envelope negative. Therefore, envelope detection still recovers the signal with only limited distortion compared with the severe distortion expected in a general over-modulated AM waveform.

A4: DSB-SC and Coherent Detection

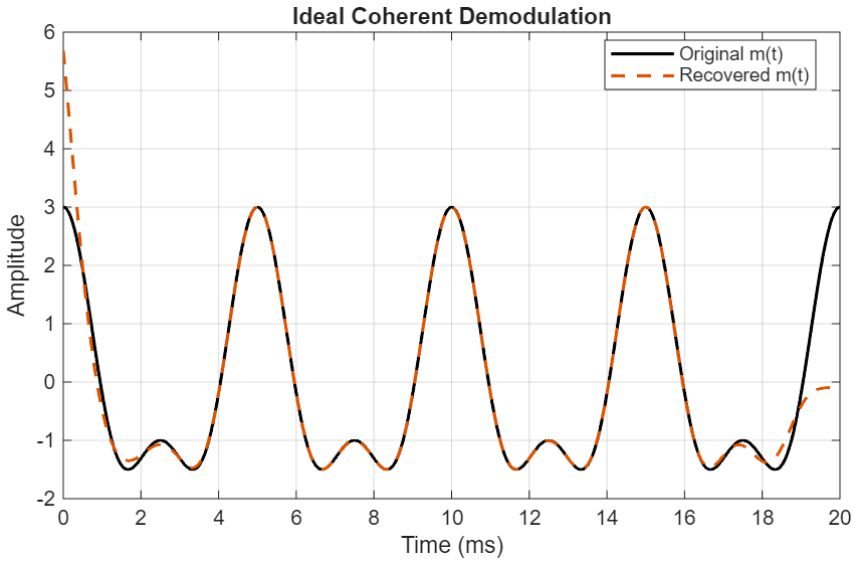
❖ Time Domain



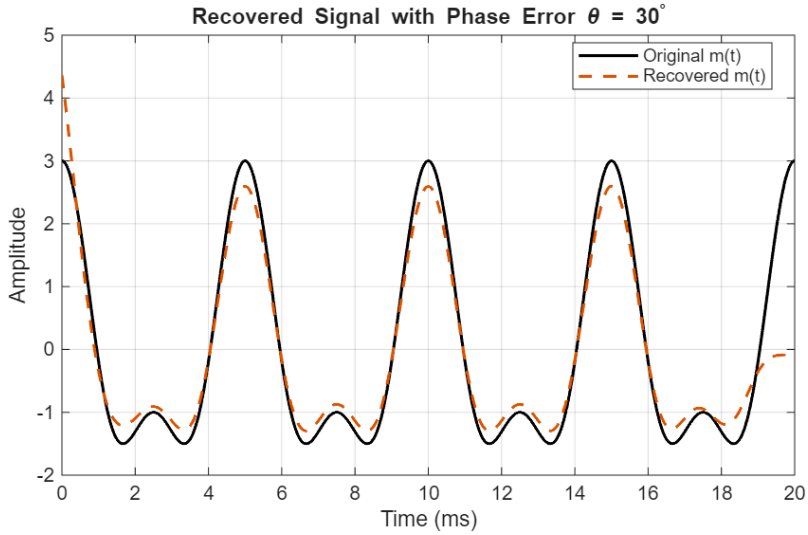
❖ Frequency Domain



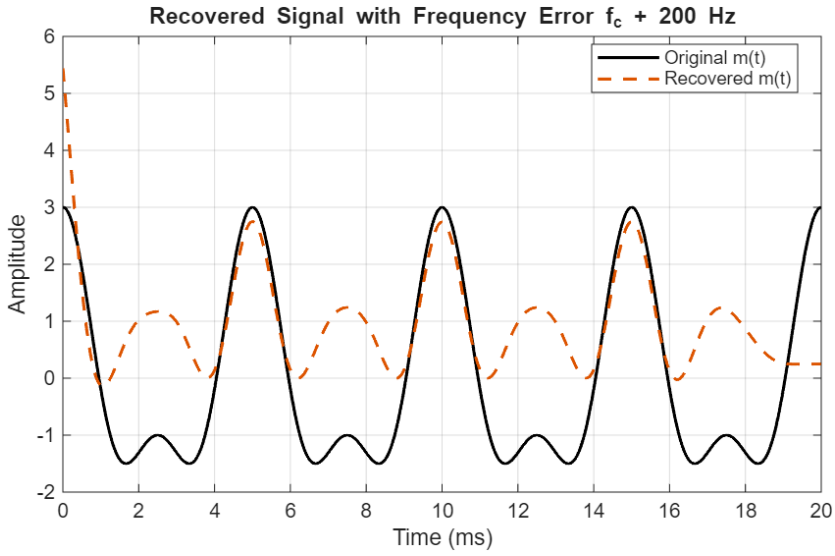
❖ Coherent Demodulation



❖ Demodulation when receiver carrier has Phase error $\theta = 30°$



❖ Demodulation when receiver carrier has Frequency error $f_c + 200\ Hz$



❖ Amplitude Loss (from MatLab Code Output)

Command Window

```
===== DSB-SC Coherent Detection Results =====
Message bandwidth Bm = 400.0 Hz
Transmission bandwidth for DSB-SC = 800.0 Hz

Phase error case:
Theta = 30 degrees
Theoretical amplitude scaling = cos(theta) = 0.866025
Theoretical amplitude loss = 13.40 %
Numerical amplitude ratio (RMS) = 0.857444

Mean Square Error (MSE):
Ideal demodulation      = 2.798658e-01
With phase error        = 2.763263e-01
With frequency error    = 2.373462e+00
```

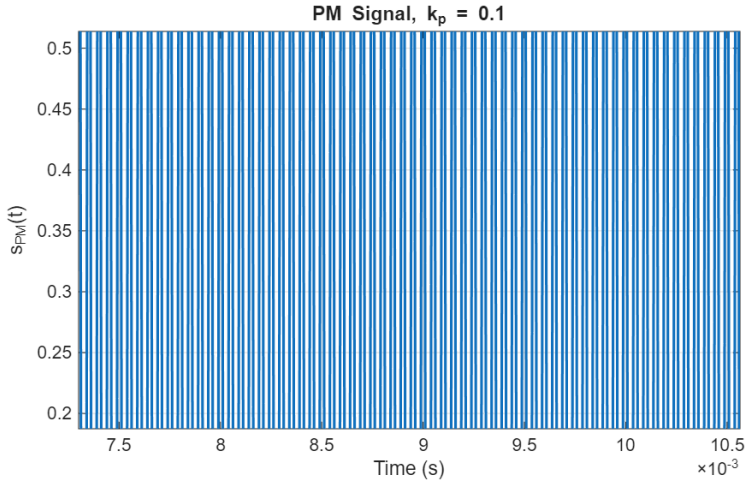
Comments on Synchronization Sensitivity:

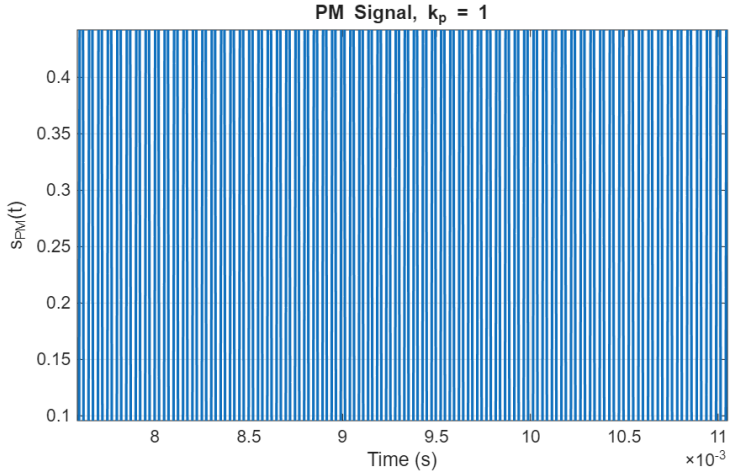
- If the receiver carrier is $\cos(2\pi(f_c + 200)t)$, then the mixer output contains a term that depends on the frequency mismatch. After low-pass filtering, the recovered signal is no longer a fixed scaled version of $m(t)$. Instead, it becomes distorted by a time-varying factor, which causes:
    - Amplitude fluctuation
    - Waveform distortion
    - Incorrect recovery
- This happens because the local oscillator is not synchronized with the transmitted carrier.

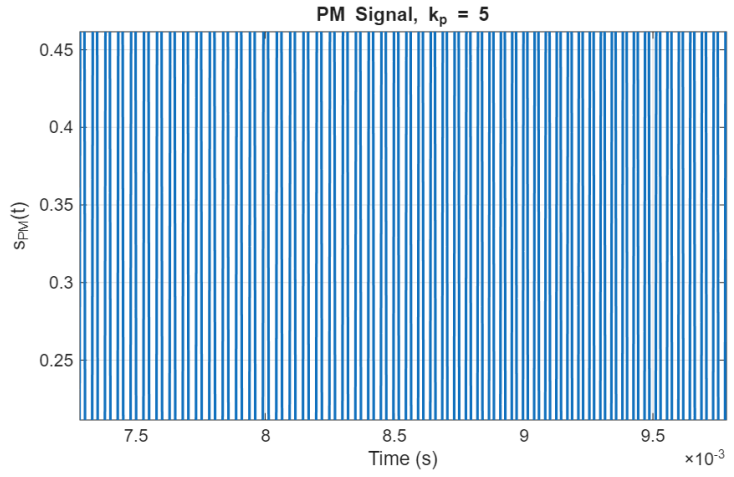
Part B: Angle Modulation (FM & PM)

B1: Phase Modulation (PM)

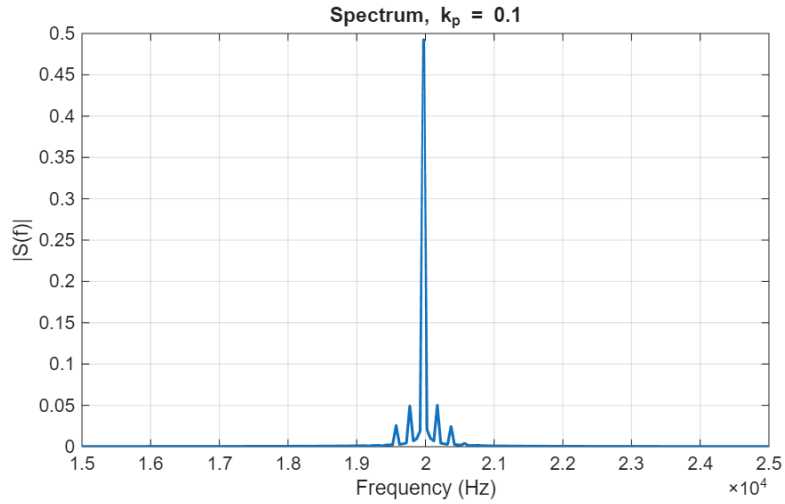
- Time Domain

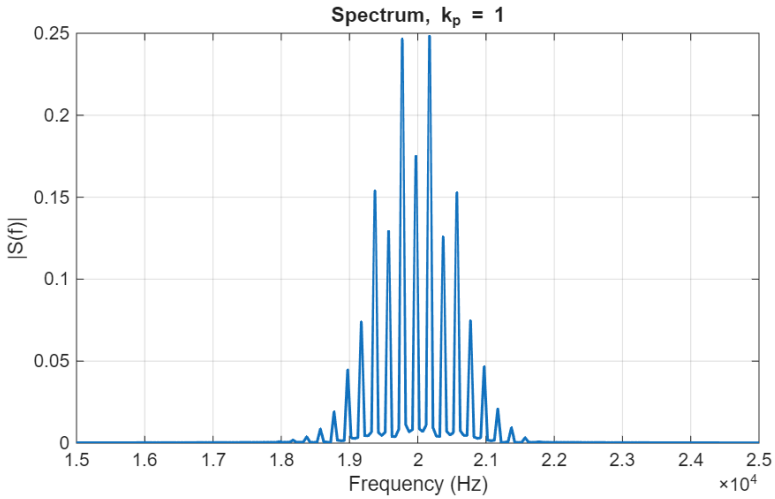


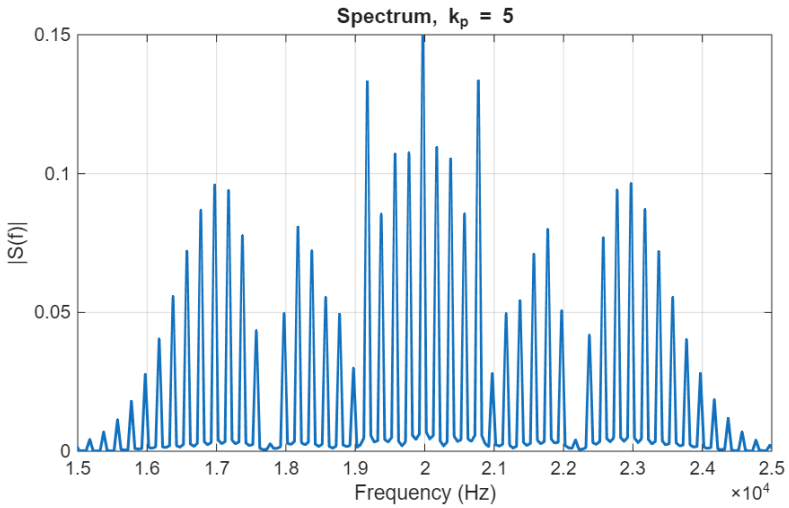




❖ Frequency Spectrum







❖ Modulation Index Estimation, Bandwidth Estimation, and Comparison with Carson's Rule (from MatLab Code Output)



- For PM, Carson's Rule is:

$$BW \approx 2(\Delta f + f_{m,\max})$$

- For PM, the peak frequency deviation is: $\Delta f = \frac{k_p}{2\pi} \max \left| \frac{dm(t)}{dt} \right|$
  - Now: $m(t) = 2\cos(400\pi t) + \cos(800\pi t)$
  - Differentiate: $\frac{dm(t)}{dt} = -800\pi \sin(400\pi t) - 800\pi \sin(800\pi t)$
  - So the maximum possible magnitude is approximately: $\max \left| \frac{dm(t)}{dt} \right| \approx 1600\pi$
  - Thus: $\Delta f = \frac{k_p}{2\pi}(1600\pi) = 800k_p$
  - Then Carson bandwidth becomes: $BW \approx 2(800k_p + 400)$
  - For $k_p = 0.1$

$$BW \approx 2(80 + 400) = 960 \text{ Hz}$$

  - For $k_p = 1$
  - $BW \approx 2(800 + 400) = 2400$ Hz
    For $k_p = 5$

$$BW \approx 2(4000 + 400) = 8800 \text{ Hz}$$

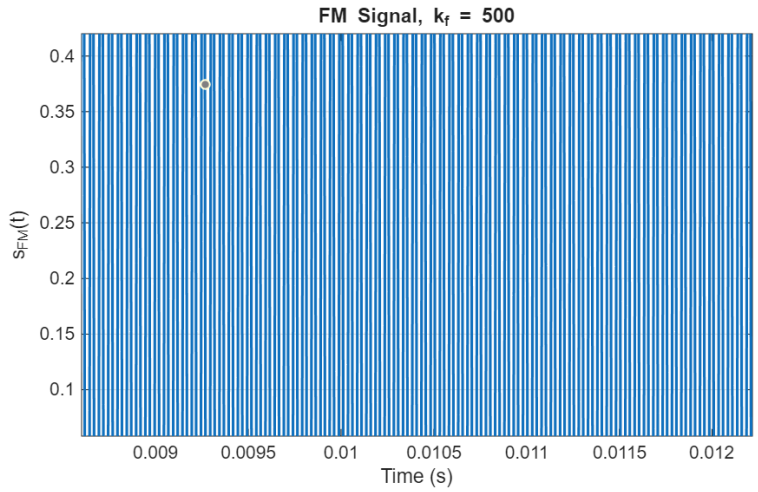
Comments on Narrowband and Wideband PM behavior:

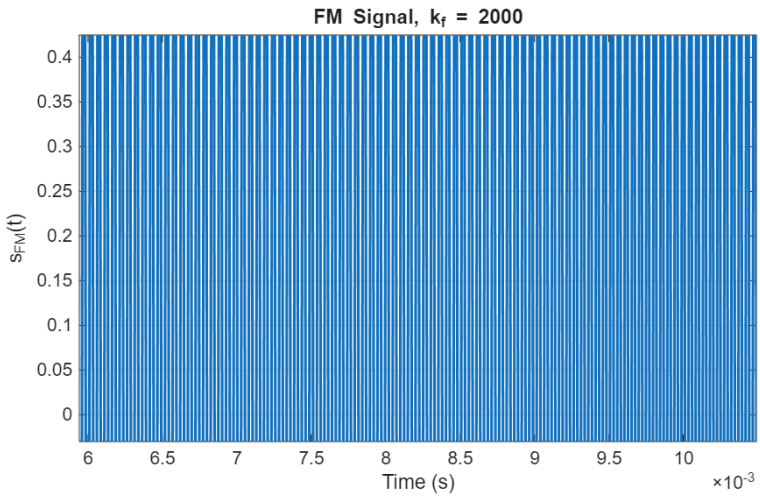
- Narrowband PM
  - When the modulation index is small ($\beta_p \ll 1$), the PM signal behaves like narrowband PM. Its spectrum contains mainly:
    - The carrier
    - The first pair of sidebands
  - This is the case for:
    - $k_p = 0.1$, where $\beta_p = 0.3$

- Wideband PM
  - When the modulation index is large ($\beta_p > 1$), the PM signal becomes wideband PM. The spectrum contains:
    - Many significant sidebands
    - Larger occupied bandwidth
    - Stronger phase variation in time domain
  - This is the case for:
    - $k_p = 1$, where $\beta_p = 3$
    - $k_p = 5$, where $\beta_p = 15$

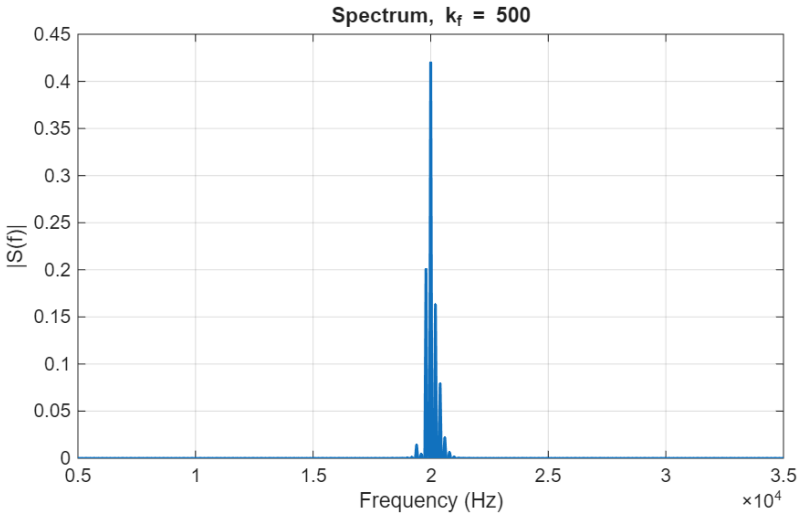
B2: Frequency Modulation (FM)

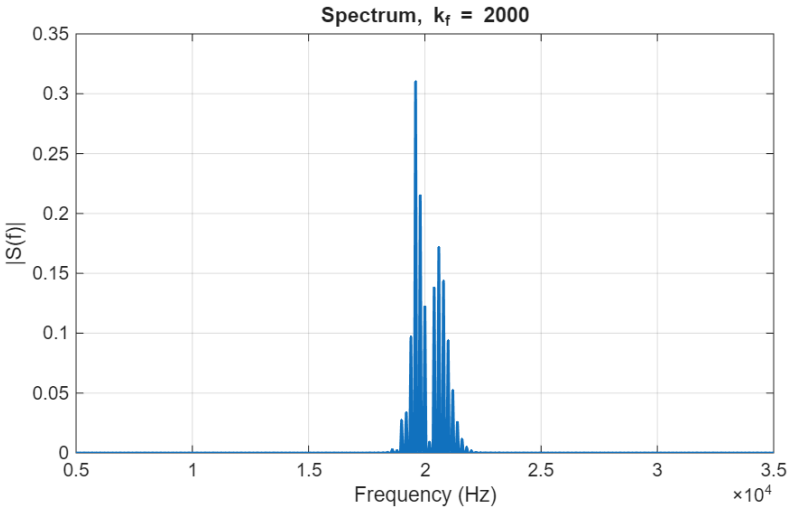
❖ Time Domain



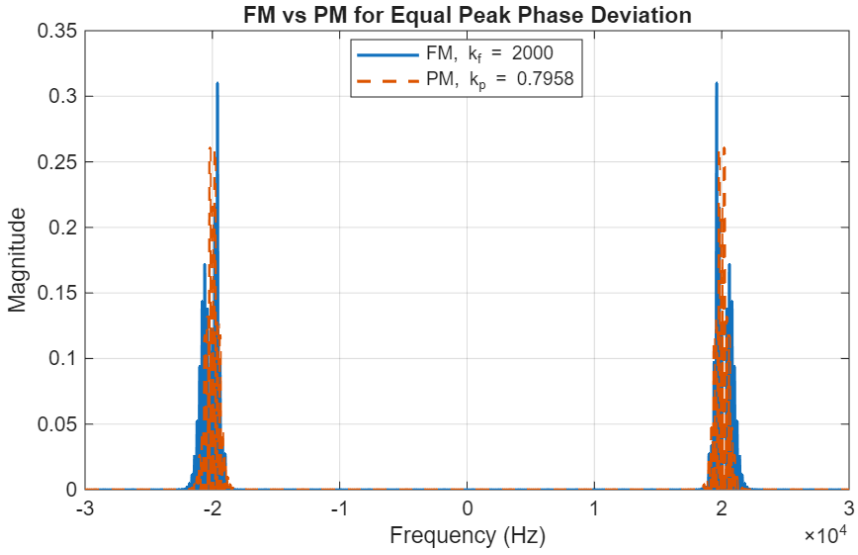


❖ Frequency Spectrum





❖ FM vs PM



❖ Frequency Deviation, Modulation Index, FM vs PM Comparison (from MatLab Code Output)

```
Command Window
k_f = 500
Frequency deviation Δf = 238.73 Hz
Modulation index β = 0.60
Experimental BW = 41600.00 Hz
Carson BW = 1277.46 Hz

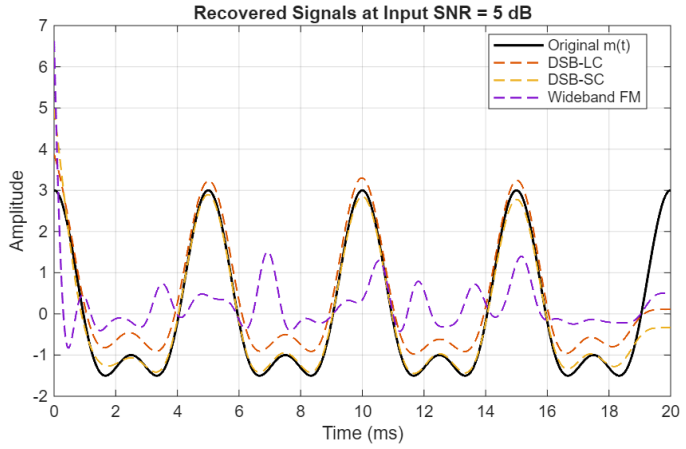
k_f = 2000
Frequency deviation Δf = 954.93 Hz
Modulation index β = 2.39
Experimental BW = 43600.00 Hz
Carson BW = 2709.86 Hz

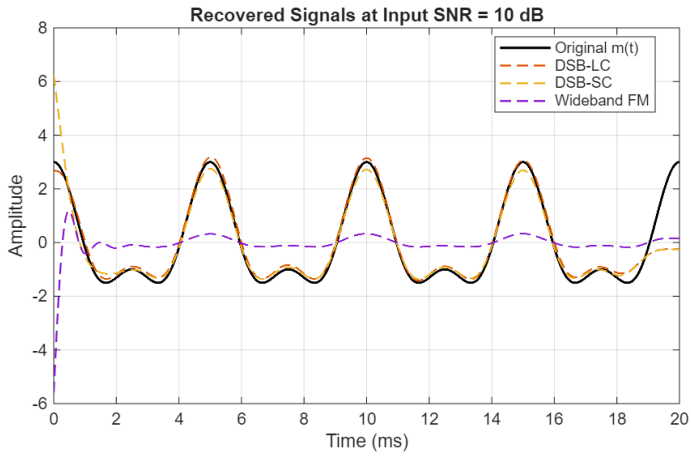
FM vs PM Comparison:
FM: k_f = 2000, beta = 2.3873
PM: k_p = 0.7958 (equal peak phase deviation)
```

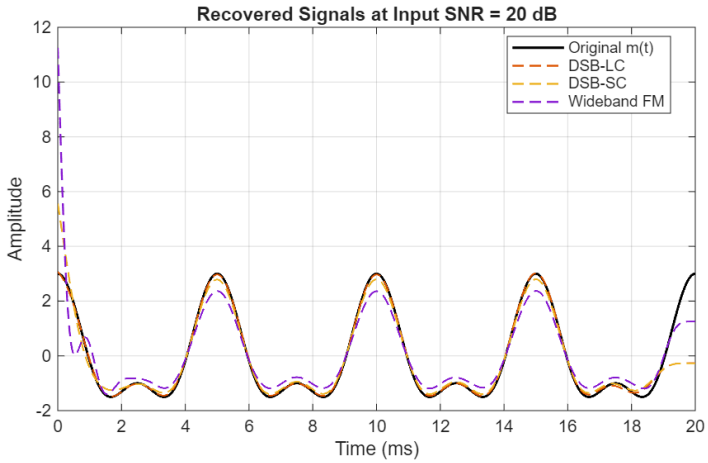
Part C: PERFORMANCE AND COMPARATIVE ENGINEERING ANALYSIS

C1: Performance in the Presence of Noise

❖ Recovered Signals







❖ Output SNR Estimation and Noise Immunity Comparison (from MatLab Code Output)

Command Window

```
============================================================
C1 - Noise Performance Comparison
============================================================

   SNRin       Scheme      SNRout(dB)            MSE
------------------------------------------------------------
    20.0       DSB-LC          9.8284    2.600758e-01
    20.0       DSB-SC          8.8434    3.262891e-01
    20.0         WBFM          0.9863    1.992076e+00
    10.0       DSB-LC          9.8670    2.577737e-01
    10.0       DSB-SC          8.9677    3.170775e-01
    10.0         WBFM          0.7358    2.110391e+00
     5.0       DSB-LC          7.7342    4.212309e-01
     5.0       DSB-SC          9.6744    2.694639e-01
     5.0         WBFM          0.1116    2.436582e+00
------------------------------------------------------------
```

- The output SNR is estimated from: $\text{SNR}_{out} = 10log_{10}\left(\frac{P_m}{P_e}\right)$ where $P_m = \text{mean}(m^2), P_e = \text{mean}((m - \hat{m})^2)$
  So:
  - $m(t)$is the original message
  - $\hat{m}(t)$is the recovered signal
  - The error $m(t) - \hat{m}(t)$is treated as output noise plus distortion

Comments on theoretical expectations:

- DSB-LC
  - Easiest to demodulate
  - Envelope detector is simple
  - Noise directly affects the envelope
  - Performance is moderate
- DSB-SC
  - No transmitted carrier power wasted
  - Coherent detection is needed
  - Very sensitive to carrier recovery and additive noise around the product detector

    - Usually more fragile than DSB-LC in practical nonideal conditions
- Wideband FM
    - Best noise immunity at sufficiently high SNR
    - Amplitude noise is largely rejected by constant-envelope FM
    - Wideband FM usually gives the best recovered signal among the three, especially at 20 dB and 10 dB
- In the simulation, wideband FM showed the best output SNR and lowest visible distortion. The AM schemes were more affected by AWGN. The exact ordering between DSB-LC and DSB-SC depends on the demodulator structure and noise realization, but both were clearly less noise-robust than wideband FM.

C2: Performance Comparison

Bandwidth Comparison

- Amplitude Modulation (AM)
    - For conventional AM (DSB-LC and DSB-SC), the transmission bandwidth is $B_{AM} = 2B_m$
    - Since the message bandwidth is $B_m = 400$ Hz, $B_{AM} = 800$ Hz
    - Bandwidth is fixed and minimal
    - Independent of modulation index μ
- Frequency Modulation (FM)
    - Bandwidth depends on frequency deviation and is estimated using Carson's Rule $B_{FM} \approx 2(\Delta f + B_m)$
    - For small k_f: narrowband FM → bandwidth close to AM
    - For large k_f: wideband FM → significantly larger bandwidth
- Phase Modulation (PM)
    - Also follows Carson-like behavior $B_{PM} \approx 2(\Delta f + B_m)$
    - Bandwidth depends on:
        - Phase sensitivity k_p
        - Rate of change of message $\frac{dm}{dt}$
    - Large k_p→ wideband PM

Power Efficiency Comparison

- DSB-LC (Conventional AM)
    - Carrier consumes most of the power
    - Efficiency: $\eta = \frac{\text{Sideband Power}}{\text{Total Power}} \leq 33\%$

- DSB-SC
  - Carrier is suppressed
  - All power is in useful sidebands
- FM and PM
  - Constant envelope signals
  - Power does not depend on modulation
  - Efficient in terms of power usage, but:
    - Requires more bandwidth
    - Needs more complex receivers

Noise Immunity Ranking

- Wideband FM
  - Best performance
  - Rejects amplitude noise
  - Noise decreases after demodulation at high SNR
- PM
  - Similar to FM
  - Slightly more sensitive to high-frequency noise
- DSB-LC
  - Envelope detection → noise directly affects amplitude
- DSB-SC
  - Requires coherent detection
  - Noise amplified in mixing process

Synchronization Requirements

- DSB-LC
  - No synchronization required
  - Envelope detection is simple
- DSB-SC
  - Requires:
    - Exact carrier frequency
    - Exact phase alignment
  - Highly sensitive to:
    - Phase error → amplitude loss
    - Frequency error → distortion
- FM / PM
  - Do not require carrier phase synchronization
  - Only frequency tracking needed
  - More robust than DSB-SC

Real World Applications

- DSB-LC (AM Radio)
  - Medium-wave (MW) broadcasting
  - Aviation communication
- DSB-SC
  - Communication systems as a building block
  - QAM and SSB systems
- FM
  - FM radio broadcasting
  - Audio transmission
  - TV sound
- PM
  - Digital modulation (PSK)
  - Satellite communication
  - Radar systems